



Reclaiming Public Spaces *for Climate Resilience* *in Maipú, Chile*

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SUMMARY

Facing increasing exposure to heat in the Sol Poniente neighborhood of Maipú, Chile, due to climate change and the urban heat island effect, local officials partnered with HOT and community leaders to map and assess their green spaces. The intervention uncovered that a third of the officially registered public squares were being misused, and 20% lacked adequate shade—putting vulnerable groups at risk during heatwaves. Equipped with accurate, community-sourced data, the municipality is now targeting investments where they're most needed, transforming neglected plots into safe, usable spaces.

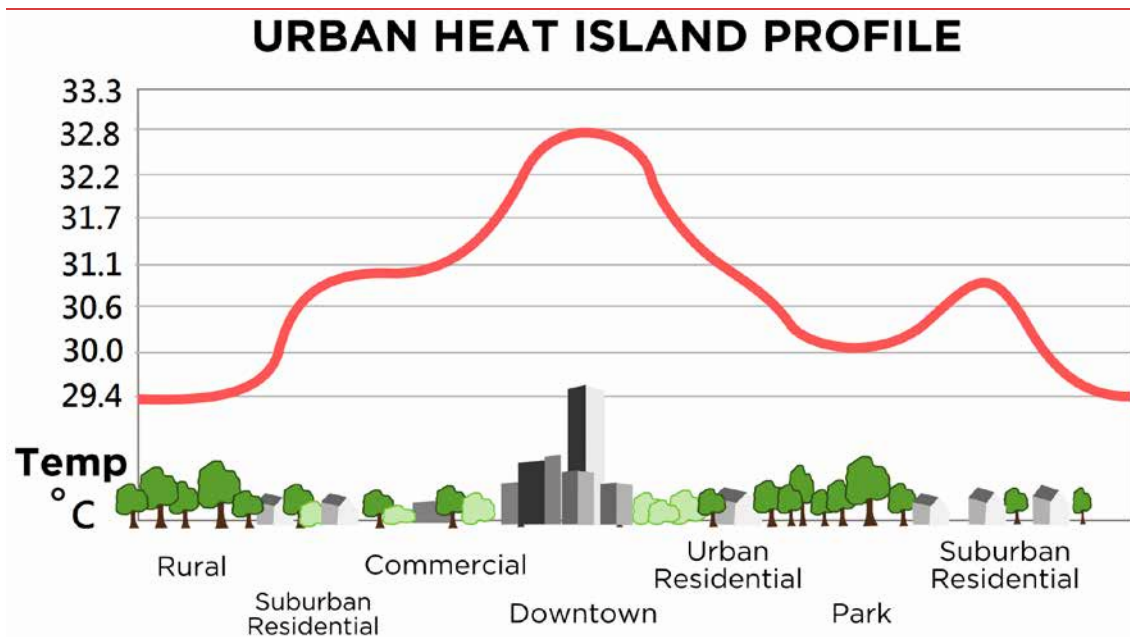
UNDERSTANDING THE *CONTEXT*

We all know it. That experience of walking on a hot summer day through a long sidewalk, the exhaust of the cars bathing us in grey, noxious gases, with not a tree in sight to provide some relieving shade or a gentle breeze to cool us down. This is known as the urban heat island effect.

The amount of concrete, the lack of permeable surfaces, and reduced vegetation are the perfect recipe for cities to get warmer than their surrounding areas ([Climate Central, 2021](#)).

Experts have pointed out that addressing the urban heat island effect in cities also works as a strategy for climate change mitigation and adaptation ([United States Environmental State Agency, 2014](#)), as increasing green spaces and reducing emissions from transportation and industries can counteract the effects of both phenomena.

But in order to address this, city governments and officials first need to spatially identify the state of their green infrastructure and use that information to determine where to invest to improve it.



This is what the municipality of Maipú, Chile, set out to do through a collaboration between the Open Government Partnership (OGP) and HOT's Latin American and the Caribbean Hub.

Beyond these issues, the members of the Planning and Development Unit of the Maipú Municipality also recognized that climate change affects the usage of these areas, especially by vulnerable populations, and can have other indirect consequences, like health issues.

“Our initial spark to join this program was a critical need to have trusted data for public investment management, especially for squares and parks. With several earned funds, we faced the challenge of investing without a clear vision of the current state of these areas, which made us look for tools that would facilitate collecting this information.”

During spring and summer, the high temperatures can render public spaces unusable, and lacking the right equipment, they can also become dangerous sites for sensitive age groups during periods of heat waves. With this in mind, the team set out to update and acquire the necessary information to invest in these public spaces and adapt them to a warmer climate.

“We see the urgency of tackling climate change in a practical and territorial way. We face a critical data vacuum about our green areas, which makes it more difficult to have effective interventions”

**Members of the
Planning and
Development
Unit of the Maipú
Municipality**



Local social leaders meeting (OGP, 2024).

OUR APPROACH IN ACTION

For this project, the Planning and Development Department of the Maipú Municipality decided to focus on the Sol Poniente neighborhood, which houses around 60,000 people. It's also where there's a higher percentage of vulnerable population (by Chile's own socioeconomic measurements) and the place they considered most lacking in open public spaces for recreation.

According to their own registry, previous to the implementation of this project, the area had 131,000 m² of green areas, which they subcategorized into squares, parks, median strips, cloverleafs, gardens, etc. The registry only showed 84 squares, but not their conditions.

Guided and supported by the CoMapper team, HOT's implementing partner, they designed a survey using KoboCollect, which included standard information like location and size, but also the conditions of the square, types of equipment present, types of use, number of people present, comfort, and safety. The survey was co-created with local social leaders, who gave input into their priorities for the areas, and who were trained to collect the information.

The resulting data—**71 polygons corresponding to Public Open Spaces**—was incorporated into the OpenStreetMap platform. Additionally, **12 plazas in the area were visually recorded using Mapillary**.



Local leader collecting field data through the survey (OGP, 2024).

OUTCOMES AND POTENTIAL IMPACT ON PEOPLE AND COMMUNITIES

After data collection and analysis, members of the local government **realized that instead of 131,000 m² of green areas, the Sol Poniente neighborhood only has 97,000 m², with 60 squares instead of 85 previously registered.**

The project made them see that, although on paper Sol Poniente had designated 85 spaces as public squares, in reality around 25 of them were being used as garbage dumpsites. They also realized that none of the green areas with playgrounds had equipment with shade, and that out of 60 public squares, 19% don't provide shade at all. This means that during warm periods, those public areas won't be used, or if they are, vulnerable users like elderly people and children are at risk of heat exposure.

This issue becomes worse by cross-referencing with maps of surface-level temperatures of the area, which show that the public spaces in the neighborhood have significantly higher temperatures than the surrounding areas of the municipality.



The municipality is currently overlapping this information with neighborhood complaints and crime databases in order to prioritize which spaces to invest in first. On the long-term, they are looking to coordinate with other Municipal Units like Community Development and the Safety Department to invest its funds in the squares with higher turnout, start adding equipment to provide shade for the users, and establish a program to adapt these public spaces to warmer temperatures.

HOW CHANGE CAN LAST

Currently, the Planning and Development Unit of the municipality is looking to expand this methodology to cover its whole area of influence beyond just the Sol Poniente neighborhood. **Francisco Medina, former Head of the Planning and Development Unit of the Municipality of Maipú, noted that the collected geographic data will be cross-referenced with information on heat islands.** This analysis will help identify priority areas for designing and implementing urban regeneration projects to facilitate adaptation to climate change, particularly in response to heat waves.

Different versions of the resulting map: physical, digital and heat map for cross-reference.

INSIGHTS FOR THE FUTURE

This project gave Maipú's local government a clear picture of its public green spaces—revealing gaps between official records and on-the-ground realities. What were once overlooked or misclassified areas are now being prioritized for investment, with a focus on creating safer, climate-resilient spaces for those most affected by extreme heat. The approach shows how combining open tools with local knowledge can lead to more responsive urban planning—offering a model for other cities facing similar challenges.



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